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标题	Copper(i) pyrimidine-2-thiolate cluster-based polymers as bifunctional visible-light-photocatalysts for chemoselective transfer hydrogenation of α , β -unsaturated carbonyls †
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摘要（原文节选）

The photoinduced chemoselective transfer hydrogenation of unsaturated carbonyls to allylic alcohols has been accomplished using cluster-based MOFs as bifunctional visible photocatalysts. Assemblies of hexanuclear clusters [Cu6(dmpymt)6] (1, Hdmpymt = 4,6-dimethylpyrimidine-2-thione) as metalloligands with CuI or (Ph3P)CuI yielded cluster-based metal organic frameworks (MOFs) {[Cu6(dmpymt)6]2[Cu2(μ -I)2]4(CuI)2}n (2), {[Cu6(dmpymt)6]2[Cu2(μ -I)2]4}n (3), respectively. Nanoparticles (NPs) of 2 and 3 served both as phot ⋯

数据表 · 表 1（7 行，含数值行 6 行）

Examples for the selective hydrogenation CO bond of cinnamaldehyde

Entry	Catalyst	Reaction conditions	Yield (%)	Selec. (%)	Ref.		
T (° C)	P H2 (bar)	t (h)					
1	Pt/YCo0.3Fe0.7O3	90	20	0.5	93.8	94.9	18
2	Ir/H – MoOx	100	20	2	93	93	19
3	Co@CN-900	80	1	48	98	99	20
4	Cu6(dmpymt)6	100	0	10	98	99	38
5a	Au/SiC	20	0	4	100	100	27
6a	3	20	0	24	94	100	—

脚注（逐字）：aReactions were promoted under visible light irradiation.

数据表 · 表 2（20 行，含数值行 20 行）

Selective photocatalytic hydrogenation of unsaturated carbonylsa

Entry	Main product	Conv.b (%)	Select.b (%)	Yieldc (%)
1		100	94	94
2		100	91	91
3		100	94	93
4		83(90)	92(91)	75(79)
5		95	93	88
6		95	95	89
7		86(92)	93(92)	78(82)
8		93	91	83
9		88	91	82
10		73(85)	90(91)	65(74)
11		90	91	85
12		84(92)	86(83)	67(72)
13		80(90)	84(82)	62(70)
14		87	84	75
15		92	87	80
16		85	86	72
17		89	84	75
18		82	86	70
19		75	92	68
20		72(85)d	85(83)d	61(68)d

脚注（逐字）：aGeneral conditions: unsaturated carbonyls (0.2 mmol), 3 (6 mg), NaOH (20 mol%), i-PrOH/MeCN (8 mL, 2/1), room temperature, blue LEDs (455 nm, 26 W), 24 h, N2.bYields determined by HPLC with biphenyl as an internal standard.cIsolated Yields.dReaction time = 36 h in parentheses.

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